

It employs Gaussian Frequency Shift Keying (GFSK) modulation technique over 2.4GHz to 2.483GHz spectrum.

It allows hands-free phone calls with a wireless cellular headset, speaker or car's audio system. It is widely used to establish a wireless connection between a mobile phone or computer and other electronic devices, including mouse, keyboard, printer, digital camera, speaker, etc.

Bluetooth 5 version offers two interfaces for the Internet of Things (IoT) devices: one for low energy operation to transmit fewer data over much longer distances, and the other for transmitting twice the data over a shorter range.

Bluetooth 5 allows users to play audio on two connected devices at the same time. People can send audio to multiple rooms in a house, create stereo effect in one space, or share audio between two sets of headphones.

Another area in which Bluetooth 5 adds improvement is Beacon technology; businesses such as retail can beam messages to nearby potential customers with offers or advertisements. People can opt out of this functionality by turning off location services and checking app permissions for retail stores.

Beacon technology can also facilitate indoor navigation, such as in an airport or shopping mall, and makes it easier for warehouses to track inventory.

Wi-Fi 6

A Wi-Fi setup consists of a wireless router that links many devices with an Internet connection depending on router configuration. Wi-Fi 6, also known as 802.11ax Wi-Fi, is the next-generation standard in Wi-Fi technology. It utilises 2.4GHz and 5GHz bands, and provides 9.6Gbps data transfer rate.

With its 1024-QAM (Quadrature amplitude modulation) technique, each symbol carries ten bits rather than eight bits, improving raw speeds by 25 per cent compared to 802.11ac 256-QAM. There are two key technologies that



NFC for payments (Credit: <https://tender-retail.acceo.com>)

improve efficiency and speed up Wi-Fi 6 connections: multi-user multiple input multiple outputs (MU-MIMO) and orthogonal frequency division multiple access (OFDMA).

If we assume a Wi-Fi connection as a series of delivery trucks delivering data packets to devices, with OFDMA each truck can deliver multiple parcels to multiple devices simultaneously, improving efficiency for both uploads and downloads. Its 8x8 MU-MIMO technique allows more than eight streams available for streaming, downloading, torrenting, playing virtual reality/augmented reality (VR/AR), massively multiplayer online games or role-playing games.

Target wake time feature allows devices to decide when and how often they will wake up to send or receive data, thus increasing device sleep time and improving battery life of mobile and IoT devices. Base service station colour marks frames from neighbouring networks to ignore interference from the neighbours' wireless networks.

Wi-Fi 6 provides communication distance of about a hundred metres, making it suitable for quality connectivity in locations with hundreds or thousands of connected devices such as stadiums and other public venues, as well as corporate networks utilising

time-sensitive, high-bandwidth and low-latency applications. Wi-Fi 6 devices operate efficiently in the most dense and dynamic connectivity settings like airports and railway stations. They provide better security features with the latest Wi-Fi Protected Access 3 (WPA3).

Near-field communication

NFC is often used for secure payment applications, fast passes, sharing information and wireless charging applications. It operates at 13.56MHz ISM frequency with a maximum range of around ten centimetres. This provides a more secure connection, which is usually encrypted. An inbuilt NFC tag allows phones, tablets, laptops and other devices to easily share data with other NFC-equipped devices.

NFC has evolved from radio-frequency identification (RFID) technology. It uses small, flat tags that can be attached to anything and used for identification, location, tracking and inventory management. When a reader unit is nearby, it transmits a high-power RF signal to the tag and reads the data stored in the memory.

NFC can be used for mobile payments and as an alternative to credit cards. It also enables transferring data between two NFC-enabled devices. With NFC, a connection is